

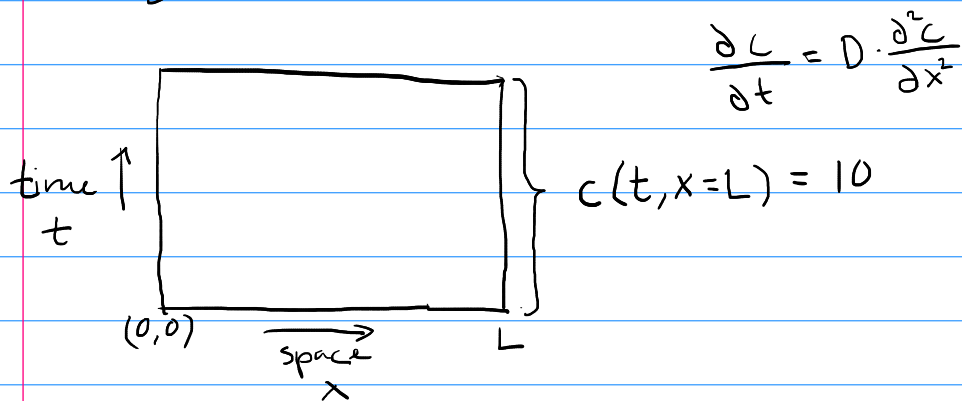
Boundary conditions

Dirichlet 1.type

Neumann 2.type

Robin 3.type

Dirichlet BC : fixed value of state variable at boundary



two options to enforce it in python

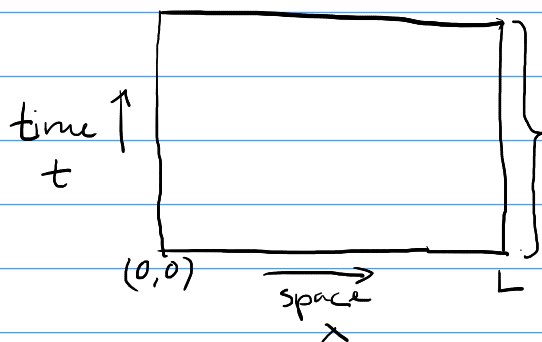
1. fix c in the initial condition → force $\frac{\partial c(t, x=L)}{\partial t} = 0$

means that c never change from its initial value at the boundary

2. set $c(t, x=L)$ directly

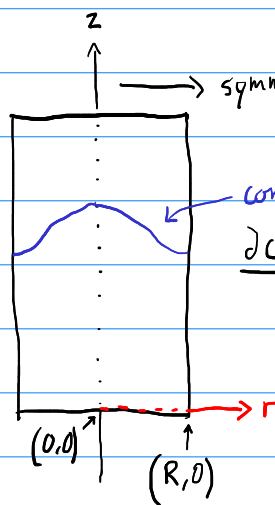
Neumann bc

when the derivative with respect to a space variable is fixed



$$\frac{\partial C(t, x=L)}{\partial x} = 0$$

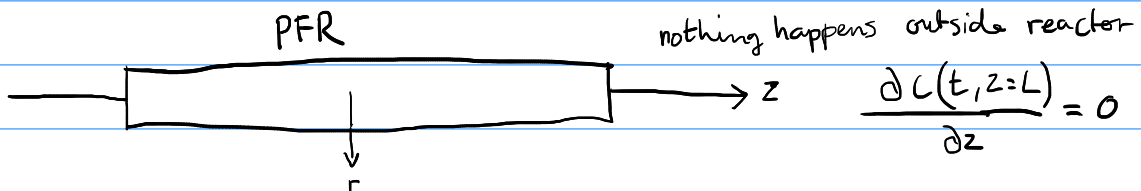
when flux is 0
closed / insulated in one end



symmetry around z

$$\frac{\partial C(r=0, z, t)}{\partial r} = 0$$

concentration profile

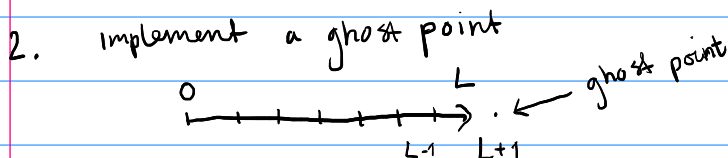


nothing happens outside reactor

$$\frac{\partial C(t, z=L)}{\partial z} = 0$$

enforce neumann bc by discretizing flux (ficks law)

$$1. \quad J = -D \frac{\partial C}{\partial x} \xrightarrow{\text{central difference}} -D \cdot \frac{C_{i+1} - C_{i-1}}{2 \cdot \Delta x}$$



$$3. \quad \text{at boundary } (x=L) \text{ flux} = 0, \text{ so } -D \cdot \frac{C_{L+1} - C_{L-1}}{2 \cdot \Delta x} = 0 \Rightarrow C_{L+1} = C_{L-1}$$

$$4. \quad \text{use } C_{L+1} = C_{L-1} \text{ in discretization of diffusion term: } \frac{\partial C}{\partial t} = D \cdot \frac{\partial^2 C}{\partial x^2}$$

$$\frac{\partial C(t, x=L)}{\partial t} = D \cdot \frac{\partial^2 C}{\partial x^2} \xrightarrow{\text{central difference}} D \cdot \frac{C_{L-1} - 2C_L + C_{L+1}}{\Delta x^2} = D \cdot \frac{2C_{L-1} - 2C_L}{\Delta x^2}$$

Robin BC

when the derivative with respect to a space variable is a function of the state variable it self

$$J = -D \frac{\partial C(t, x=L)}{\partial x} = f(C_L) = \underbrace{k_m \cdot (C_L - C_{x>L})}_{\text{mass transfer by convection}} \quad \swarrow \text{eg bulk air or bulk water outside domain}$$

$$q = -k \frac{\partial T(t, x=L)}{\partial x} = f(T_L) = \underbrace{h \cdot (T_L - T_{x>L})}_{\text{heat transfer by convection}} \quad \swarrow \text{temperature outside domain}$$

$$J = -D \frac{\partial C(t, x=L)}{\partial x} = f(C_L) = k \cdot C_L^{\alpha}$$

reaction at boundary
balance the flux

$$J = -D \frac{\partial C(t, x=L)}{\partial x} = f(C_L) = k_m \cdot (C_L - C_{Air})$$

↓
isolate → $C_{L+1} - C_{L-1} = k_m \cdot (C_L - C_{Air})$

$$C_{L+1} - C_{L-1} = \frac{2\Delta x \cdot k_m}{-D} \cdot (C_L - C_{Air})$$

$$C_{L+1} = \frac{2\Delta x \cdot k_m}{-D} \cdot (C_L - C_{Air}) + C_{L-1}$$

substitute into diffusion term

$$\frac{\partial^2 C}{\partial x^2} = \frac{C_{L+1} - 2 \cdot C_L + C_{L-1}}{\Delta x^2} = \frac{\frac{2\Delta x \cdot k_m}{-D} \cdot (C_L - C_{Air}) + C_{L-1} - 2C_L + C_{L-1}}{\Delta x^2}$$